PROJECT TITLE | DATA SCIENCE / AI

AI Interview Pro

An AI-powered interview practice platform with personalized feedback and learning recommendations.

Category	Project type	Intended users
Data Science / AI	Full-stack web application	Students and job seekers

Project overview

InterviewPro helps candidates practice technical, behavioral, aptitude, resume-based, and communication interviews. It saves practice sessions and feedback so learners can track progress and choose a useful next step. Scores are for practice guidance, not predictions of hiring outcomes.

Problem and solution

Candidates often lack realistic interview practice, timely feedback, and a clear improvement path. This project combines structured mock interviews, transparent rubric-based feedback, progress analytics, resume-aware questions, and personalized recommendations grounded in the learner's recorded activity.

Project plan

PHASE	ACTIVITIES AND DELIVERABLES
1. Requirements and design	Confirm user journeys, interview modes, privacy needs, data fields, and responsive dashboard flows.
2. Core application	Build account access, authenticated dashboard, interview setup, question flow, and answer submission.
3. Feedback and personalization	Save sessions and reports; calculate transparent rubric feedback; show history, analytics, skills, and evidence-based learning steps.
4. Quality and deployment	Test frontend and API flows; use managed PostgreSQL; secure resume uploads and secrets; deploy the frontend and API over HTTPS.

IMPLEMENTATION IDEA

How the platform works

A web client makes authenticated requests to a Python API. The API retrieves only the signed-in user's records, evaluates answers, stores reports, and returns dashboard data.

1. Practice Select an interview mode, role, difficulty, and optional resume.	**2. Evaluate** Assess answers with transparent rubric criteria and topic evidence.
3. Save Persist questions, responses, scores, and completion status for the user.	**4. Improve** Display progress and a learning plan based on recorded results.

AI and data approach

Interview questions and optional resume context guide practice. Answer feedback combines rubric checks with a lexical similarity baseline. The learning coach identifies weak areas only when completed scored answers provide sufficient evidence; otherwise, it suggests a short diagnostic. Optional hosted AI can enhance a study plan only with user consent. Personal data is scoped to the authenticated account.

Tools and technologies

Frontend	React, TypeScript, Vite, HTML, CSS
Backend and API	Python, FastAPI, REST, Pydantic
Database	PostgreSQL for production; SQLAlchemy ORM; Alembic migrations; SQLite for local development
AI and evaluation	Scikit-learn TF-IDF baseline, rubric-based feedback, optional consent-based OpenAI-compatible plan enhancement
Security and deployment	JWT authentication, HTTPS, environment-managed secrets, Git, Vercel frontend, Render API and database

Expected result and evaluation

Deliver a responsive, authenticated practice platform with saved history, answer feedback, resume-aware context, dashboard analytics, and personalized learning steps. Evaluate with API and UI tests, verify per-user data isolation, and conduct staged load testing before scaling up.

Core principle
Recommendations must be supported by actual practice evidence. When evidence is limited, guide the learner to a diagnostic session instead of inventing a weakness.